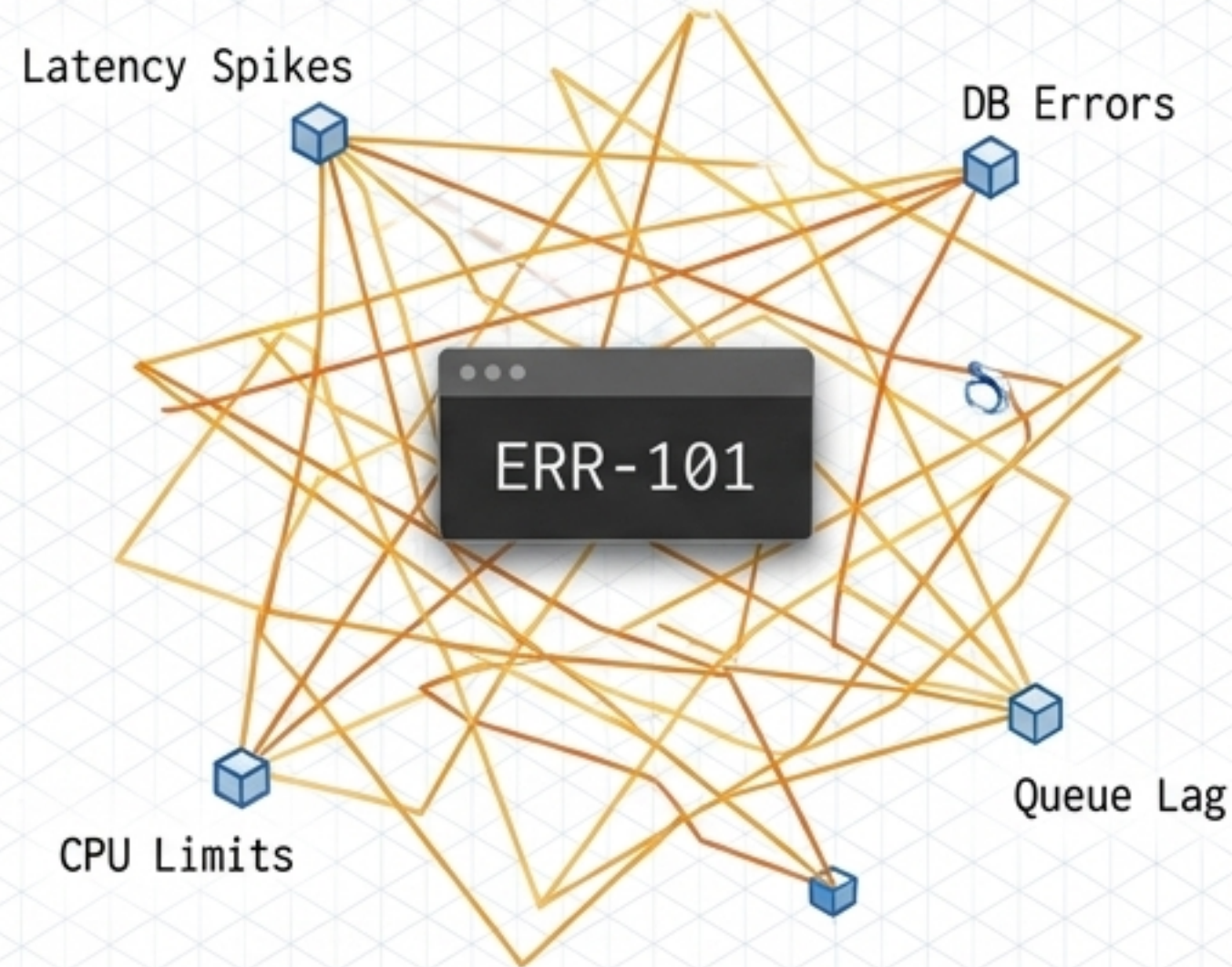


SentryIQ Architectural Blueprint

Engineering an Explainable, Event-Driven Incident Intelligence Platform for Enterprise Banking.

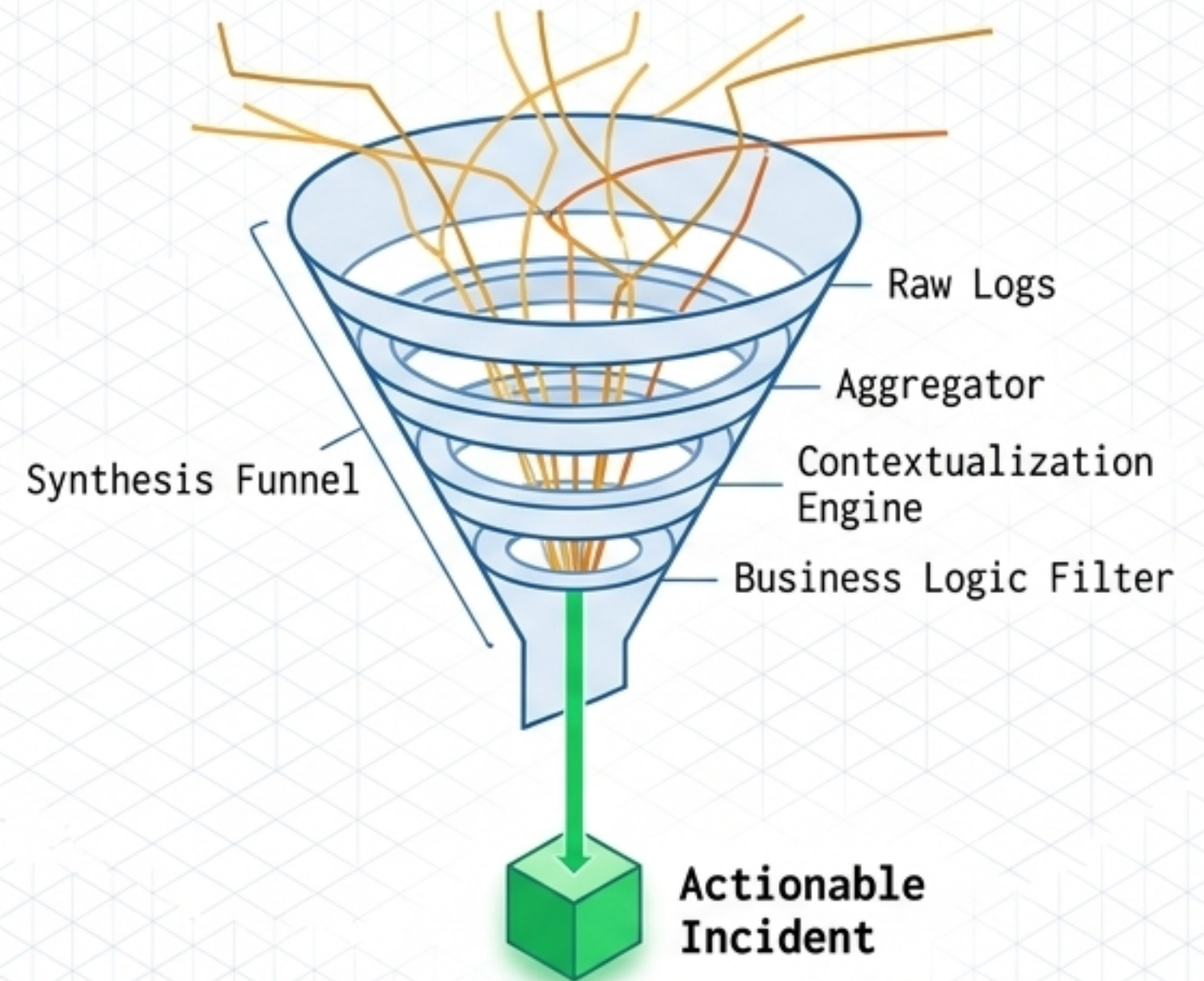
The Core Challenge: Alert Fatigue vs. Actionable Intelligence

The Challenge: Alert Fatigue



Modern banking systems generate millions of fragmented raw logs. Without business context, operations teams face alert fatigue, leading to higher Mean Time to Detect (MTTD) and Resolve (MTTR).

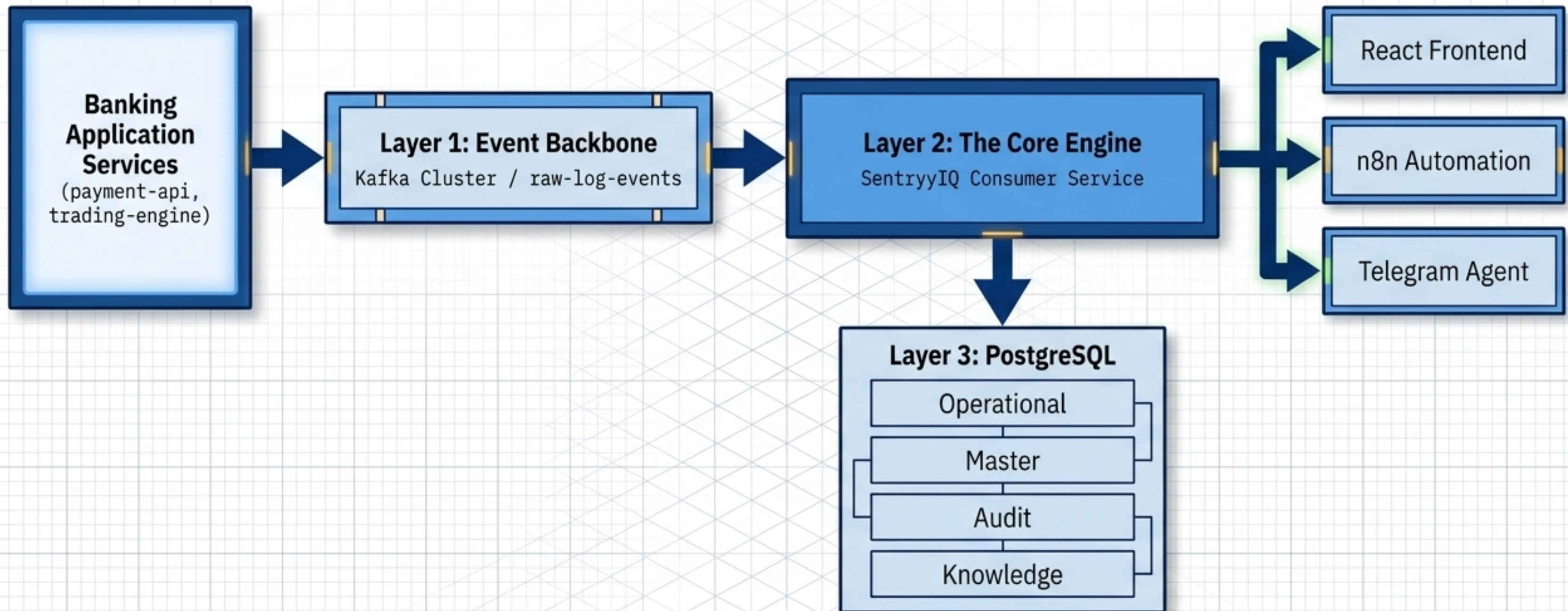
The Solution: Actionable Intelligence



SentryIQ enforces a Service-Level Aggregation model, synthesizing thousands of technical endpoints into a single, explainable measure of service health and quantifiable business risk.

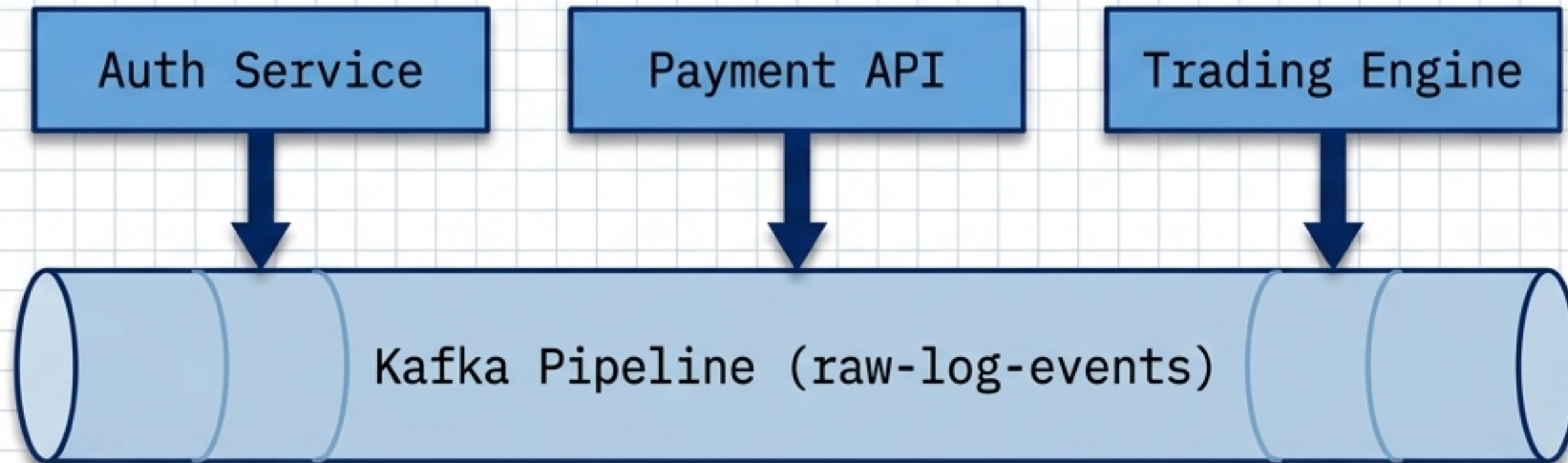
SentryyIQ System Architecture (C4 Container View)

An end-to-end event-driven pipeline designed to ingest decoupled, high-throughput telemetry and output actionable, agentic workflows.



Layer 1: Decoupled Telemetry Ingestion (Event Backbone)

Kafka buffers telemetry for 3 days. This enables horizontal scalability, replay capabilities for model back-testing, and complete decoupling of producers from consumers without acting as long-term storage.



14-Field Raw Event Schema

Routing: timestamp, service, endpoint, region

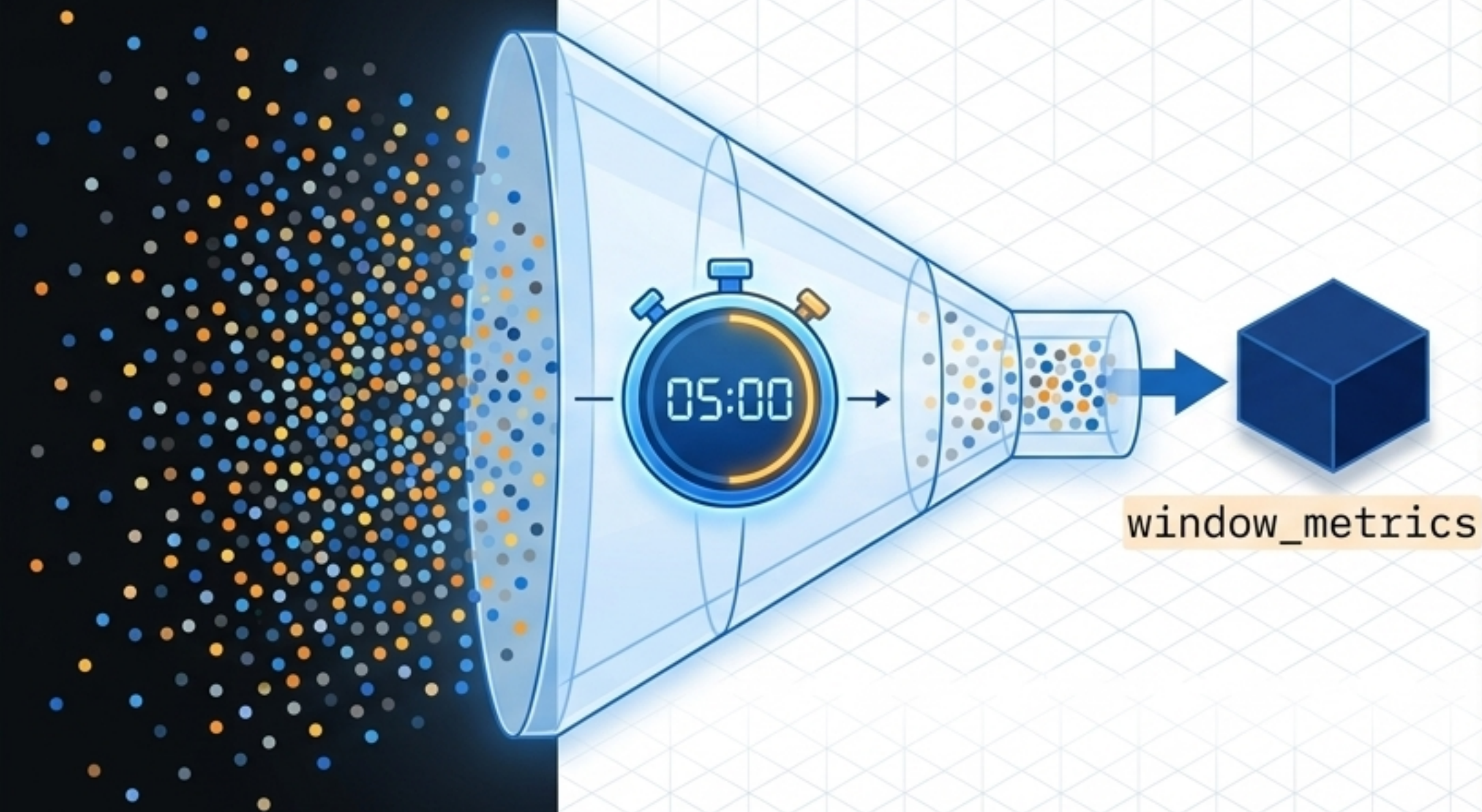
Impact Identifiers: client_id, machine_id

Performance: latency_ms, cpu_usage, queue_lag

Metadata: error_code, severity, status

Layer 2: The One Window Aggregation Strategy

Synthesis Funnel

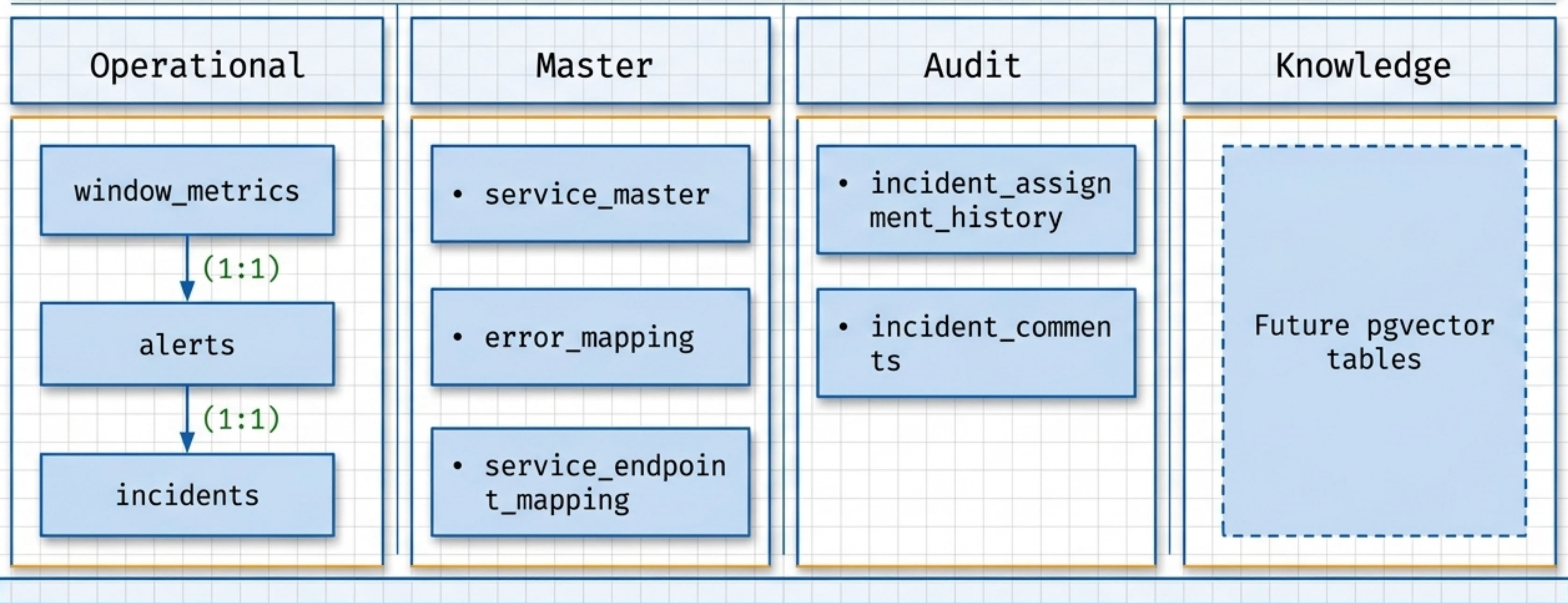


ADR-01: Service-Level Aggregation

- **Decision:** Group incoming Kafka records strictly by service, `window_start`, and `window_end` (e.g., `payment-api`, `10:00-10:05`).
- **Operational Value:** Prevents alert storms. Thousands of endpoint-level logs are synthesized into into **exactly one actionable 5-minute metric block**. Reduces computational overhead while identifying systemic service degradation.

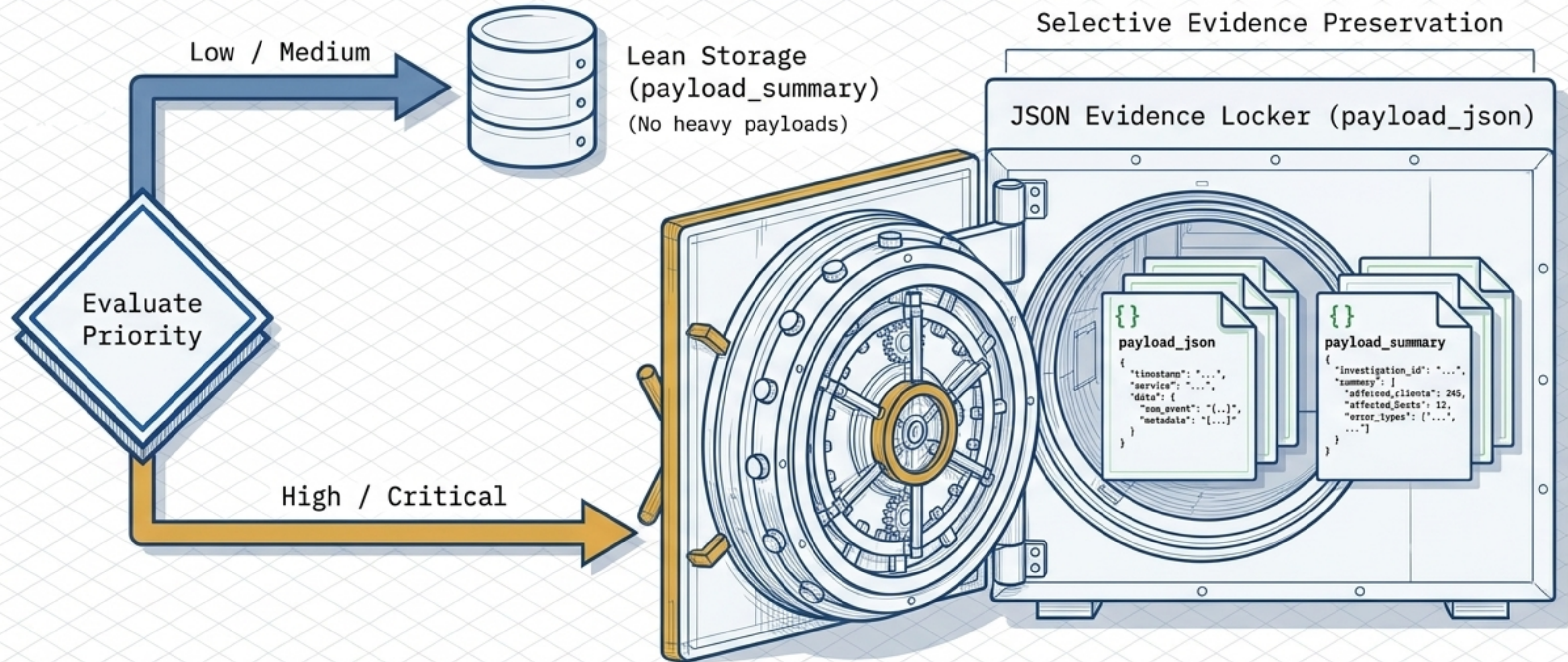
Layer 3: System of Record (The Four-Pillar Schema)

Strict separation of operational data from business metadata. This ensures 1:1:1 accountability (One Window -> One Alert -> One Incident) and allows rapid AI context-building.



Payload Intelligence & Selective Preservation

The Storage Dilemma: Storing full telemetry as flat columns for billions of successful banking transactions is cost-prohibitive. SentryIQ dynamically captures deep forensic evidence exclusively for major outages.



ML Model Selection Matrix

Model	Precision	Recall	F1-Score	Operational Stability
Isolation Forest (Winner) 🏆	1.0000	0.8105	0.8953	High - robust to telemetry variability
One-Class SVM	0.1523	0.7053	0.2505	Low - boundary instability
Local Outlier Factor (LOF)	0.0130	0.0947	0.0229	Fail - distorted by temporal aggregation

Architectural Conclusion: Tree-based Isolation Forest preserves anomaly separability effectively without generating the false-positive alert noise typical in banking environments.

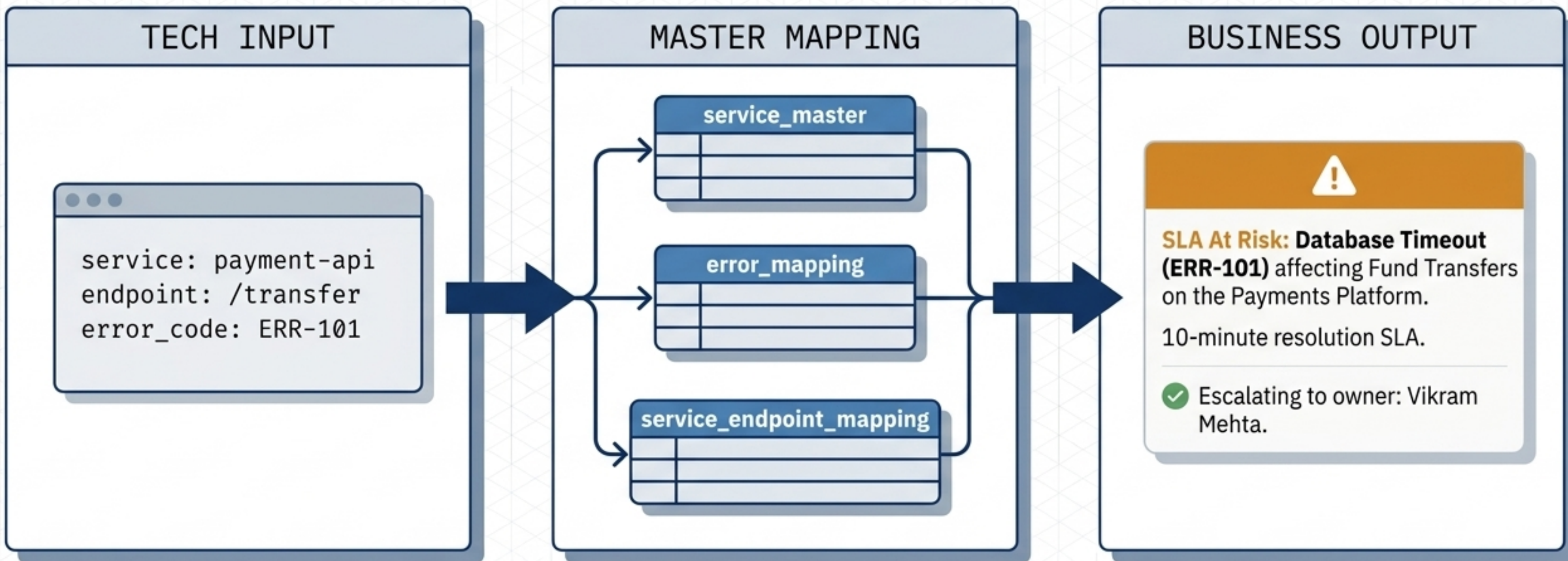
Deep Dive: The Explainable Ensemble Risk Engine

The Explainability Mandate: No black-box AI. Every generated alert permanently records its constituent sub-scores, allowing **operators to trace exact root causes**.



Bridging the Tech-to-Business Gap

SentryyIQ acts as a Decision-Support Layer. Master tables provide immediate translation, removing the need for tribal knowledge during critical incidents.



Advanced Impact Analysis

Infrastructure Hotspots



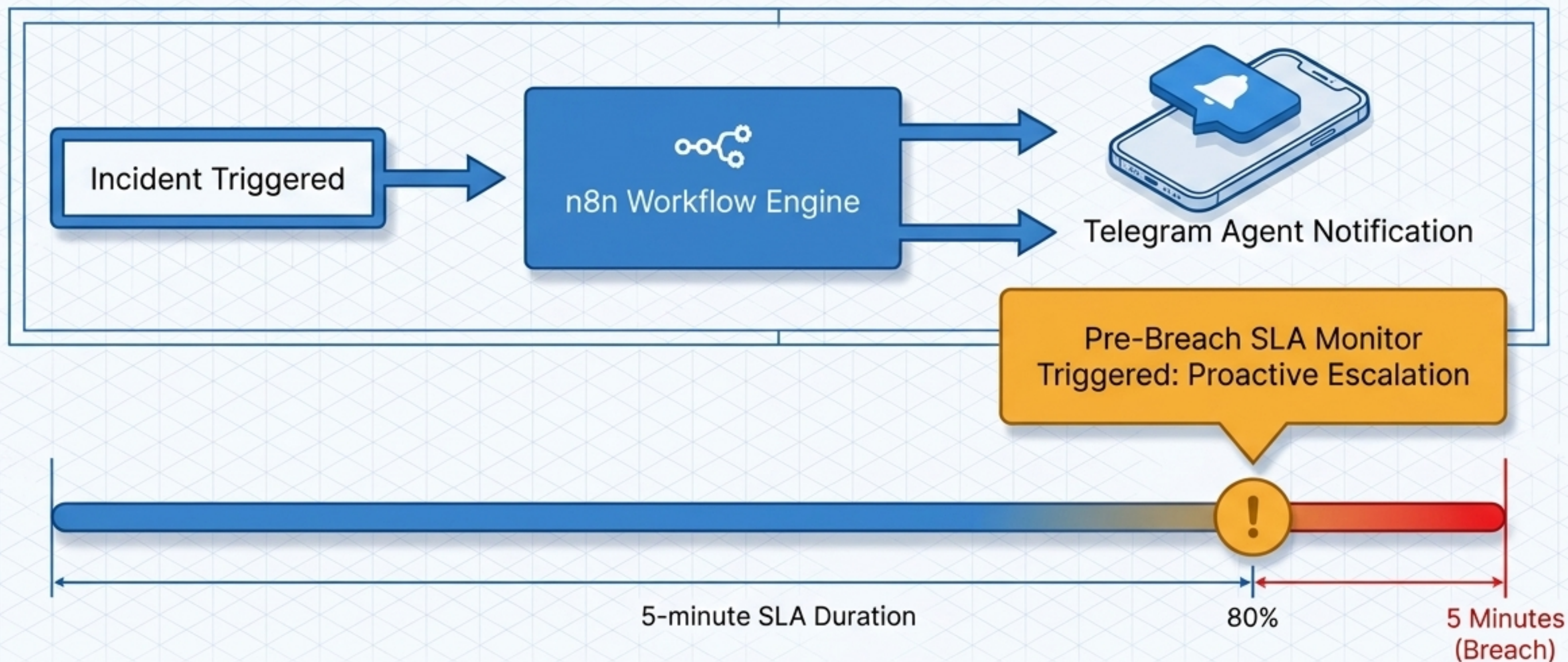
Distinguishes between systemic code bugs (requiring service restart) and localized container failures (high ML score + low affected hosts). Pinpoints exactly which node to kill.

Customer Impact



Aggregates unique client_id counts. Identifies unlucky clients failing across multiple services (e.g., failed login AND failed withdrawal) to enable proactive success outreach before they complain.

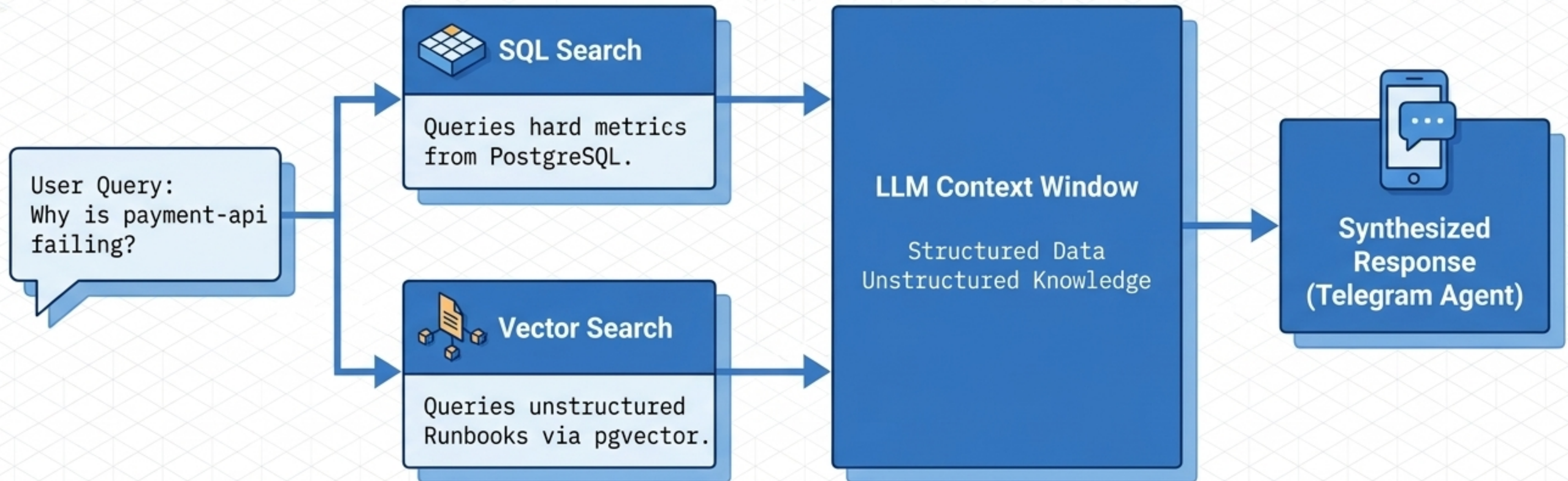
ACTION, AUTOMATION, AND GOVERNANCE



Shifting from reactive reporting to proactive escalation. SentryIQ triggers interventions at 80% SLA duration, delivering executive briefings directly to mobile devices with zero context switching.

Future Architecture: Hybrid Enterprise RAG

Preventing AI hallucinations. By grounding responses in real-time PostgreSQL facts alongside vector-embedded runbooks, SentryyIQ enables highly accurate natural language investigations.



Deployment Architecture: Target Cloud Topology

Cloud-native by design. The containerized architecture transitions seamlessly to managed cloud services requiring zero architectural redesign, ensuring enterprise readiness.

